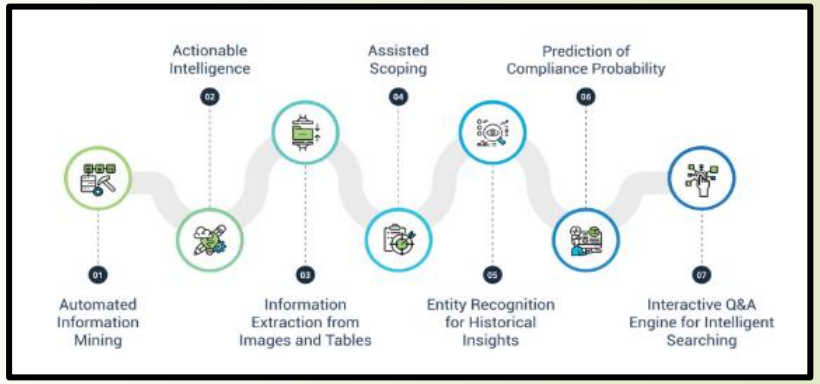
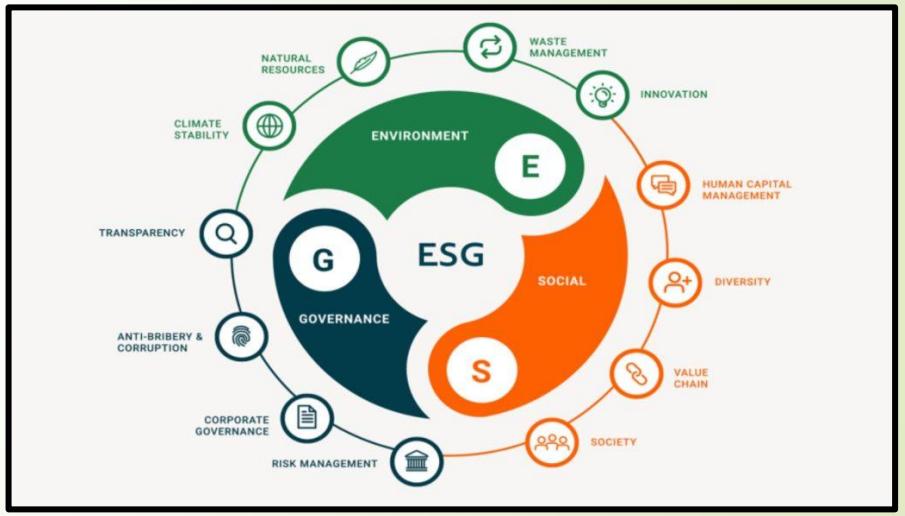
INTRODUCTION TO ESG REPORTING

- ESG reporting focuses on environmental, social, and governance factors for transparency in sustainability.
- Growing demand for effective and timely ESG reporting due to regulatory pressure.
- Traditional methods are error-prone, time-consuming, and incapable of real-time insights.
- AI transforms ESG reporting through automation, real-time insights, and reduced human involvement.
- AI integration ensures more efficient ESG reporting, enhancing compliance and strategic alignment.



PROBLEM DEFINITION

- Traditional ESG reporting is manual, slow, and error-prone, leading to delays and inaccurate disclosures.
- Lack of standardized, automated tools hinders small businesses' ability to meet regulatory demands.
- Growing demand for precise and timely ESG disclosures makes real-time data integration essential.
- The absence of effective ESG reporting systems creates barriers for small businesses and VSMEs.
- AI helps by automating data collection, reducing errors, and ensuring timely, accurate ESG reporting.



RESEARCH AIM AND OBJECTIVES

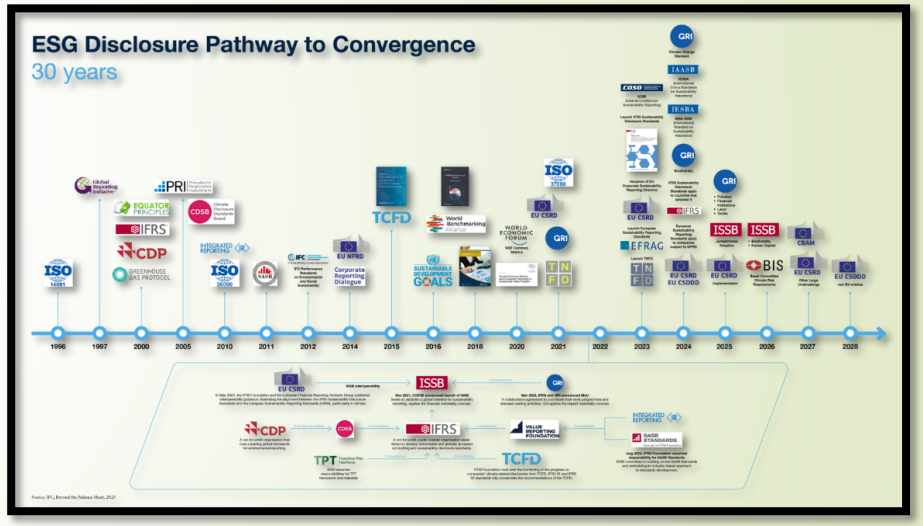
- Aim: Develop an AI-powered ESG reporting tool integrating automation, real-time data, and VSME reporting.
- To Award Key Features Required of an AI-Assisted ESG Reporting Tool
- To Implement a Structure of Inclusion of VSME Reporting in the Tool
- To Determine the Effect of AI Integration on ESG Reporting Precision
- To Create a User Interface for Streamlined ESG Report Creation

RESEARCH QUESTIONS

- What AI technologies and approaches can streamline ESG reporting through automation, error minimization, and accuracy?
- Which is the most important feature, such as automation, real-time data integration, or user interface design, that is required for an effective ESG reporting tool?
- How does the VSME reporting system enable automation of ESG reporting and improve reporting efficiency?
- What are the technical, data-related, and ethical challenges in developing AI models for analyzing and reporting ESG data?
- How can AI-enabled tools enhance ESG compliance and actionable insights for informed decision-making in organizations?

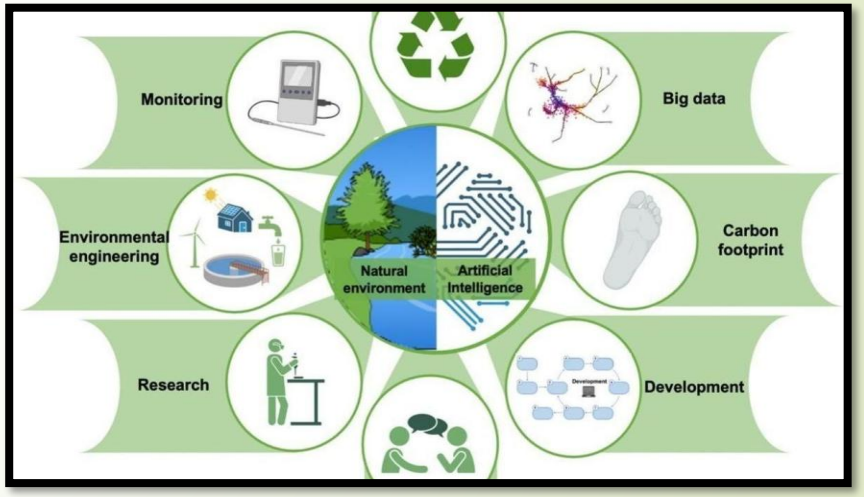
LITERATURE REVIEW: EVOLUTION OF ESG REPORTING STANDARDS AND FRAMEWORKS

- ESG reporting has evolved from voluntary disclosures to standardized frameworks for transparency.
- Global ESG frameworks like GRI, SASB, and TCFD help standardize sustainability disclosures.
- Regulatory frameworks like CSRD and ESRS ensure more detailed and uniform ESG reporting.
- VSMEs face challenges due to limited resources and difficulty in meeting complex ESG standards.
- AI-driven solutions automate ESG data collection, providing real-time insights for improved compliance.



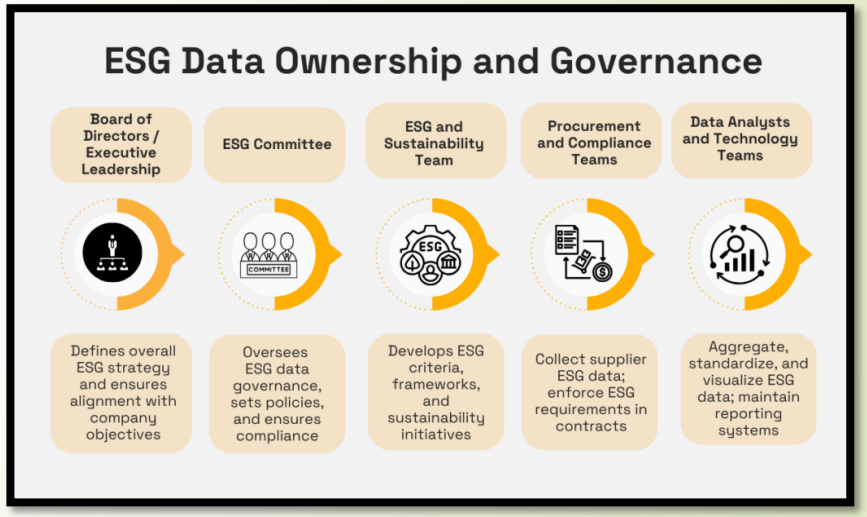
ROLE OF ARTIFICIAL INTELLIGENCE IN SUSTAINABILITY AND COMPLIANCE

- AI automates data collection, analysis, and report generation, improving ESG reporting accuracy.
- AI models enable real-time monitoring of sustainability performance and compliance with regulations.
- AI tools identify risks and provide predictive analytics for better decision-making.
- AI helps detect greenwashing by analyzing ESG disclosures for inconsistencies and misstatements.
- AI offers scalable solutions for VSMEs to implement ESG reporting tools with minimal capital.



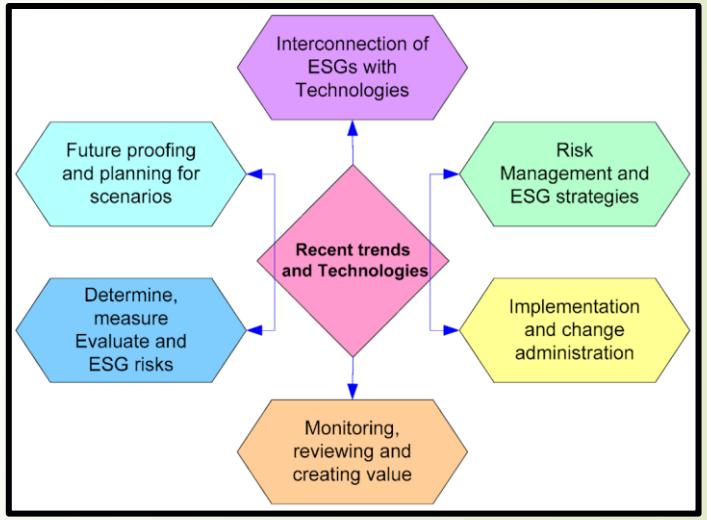
AUTOMATED DATA COLLECTION AND ESG METRIC EXTRACTION

- APIs, RPA, and IoT sensors automate real-time collection of ESG data such as energy use and emissions.
- APIs enable seamless data exchange between internal systems and external providers for accurate reporting.
- RPA automates repetitive tasks, reducing human errors in data entry and report aggregation.
- IoT sensors track sustainability metrics in real time, supporting compliance and operational decisions.
- Web scraping and OCR technologies ensure comprehensive data coverage for automated ESG reporting.



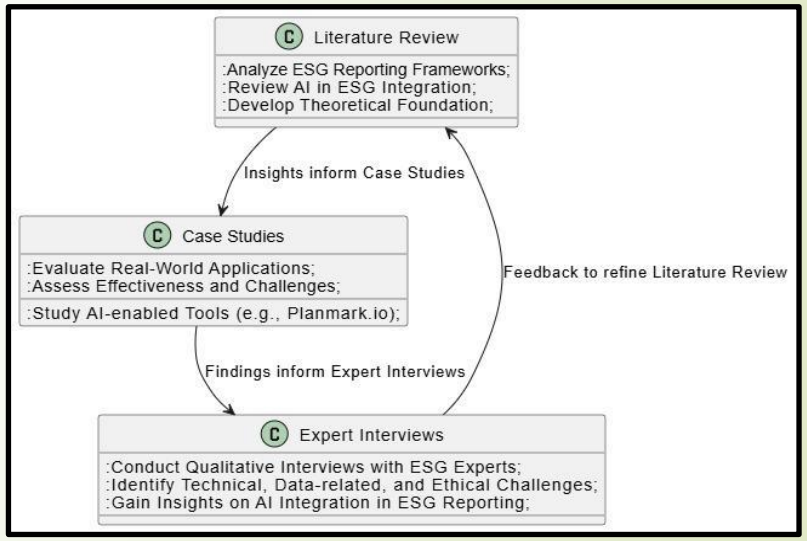
NATURAL LANGUAGE PROCESSING FOR ESG REPORT GENERATION

- NLP automates ESG report generation by converting structured data into narrative form.
- LLMs like GPT generate human-like text for ESG disclosures, ensuring consistency and quality.
- NLP reduces reliance on manual editing, improving report consistency and turnaround time.
- NLP-based systems adjust language to suit different stakeholders (investors, regulators, management).
- AI tools enable businesses to generate standardized, high-quality ESG reports without specialized expertise.



RESEARCH METHODOLOGY: DATA COLLECTION

- Secondary data is collected through a literature review of ESG frameworks, AI applications, and case studies.
- Primary data includes case studies of businesses using AI-powered ESG tools like Planmark.io.
- Expert interviews with ESG professionals, AI developers, and industry experts gather insights on challenges.
- Qualitative analysis of interviews and case studies helps evaluate the impact of AI on ESG reporting.
- Data from primary and secondary sources will inform the development of AI-based ESG reporting solutions.



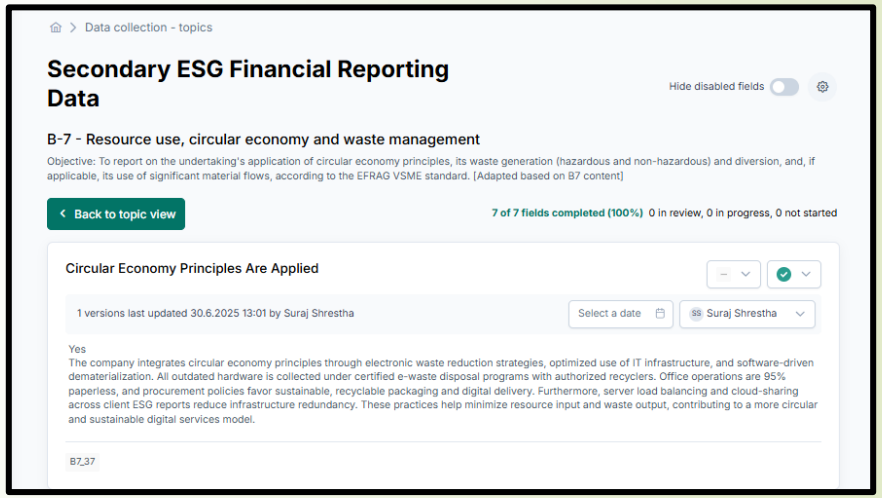
RESEARCH FINDINGS: AI IN ESG REPORTING - PART 1

- AI-driven ESG reporting system increases transparency and compliance with sustainability by automating data collection, risk detection.
- The AI system helps companies monitor performance, close ESG gaps, and align strategies with environmental.
- The introduction of AI in ESG reporting addresses the growing regulatory pressure and demand for timely, factual.
- The tool offers a user-friendly interface, allowing small enterprises to manage ESG data, track performance.



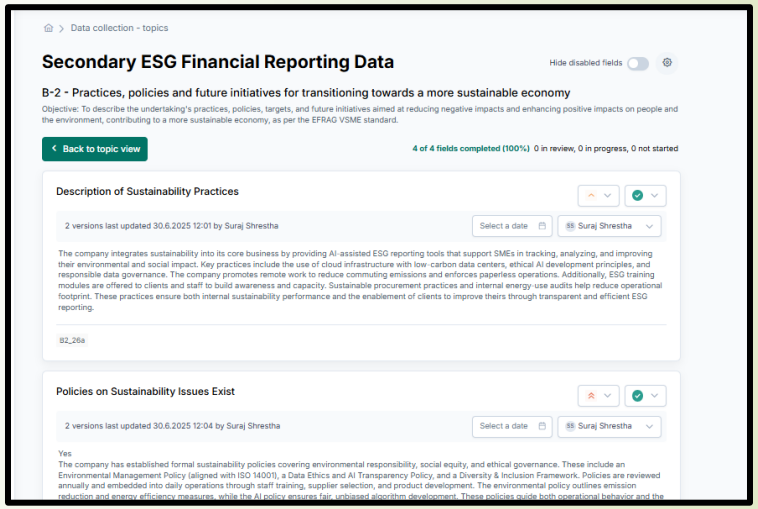
RESEARCH FINDINGS: AI IN ESG REPORTING - PART 2

- The AI-driven system automates the data collection process by integrating structured datasets, such as ESG and financial metrics, from various sources for analysis and reporting.
- Data Collection Window allows users to access, manage, and update ESG financial datasets in real time, ensuring efficient, cross-industry, and cross-sustainability measures for reporting.
- AI systems are capable of analyzing large datasets on environmental, social, and governance factors such as energy consumption, waste management, and workforce diversity to generate accurate, actionable insights.
- The system includes a feature for monitoring and managing ESG reports' basic sensitive information to ensure compliance with data security standards and regulatory provisions.



RESEARCH FINDINGS: AI IN ESG REPORTING - PART 3

- Key findings indicate that AI-driven ESG reporting systems improve waste management practices, with 92.7% of non-hazardous waste being diverted for recycling and reuse, promoting circular economy principles.
- The tool tracks environmental performance metrics such as GHG emissions, energy consumption, and water usage, helping businesses reduce their ecological footprint and align with sustainability goals.
- Workforce-related insights include gender balance and turnover rates, with AI monitoring retention, diversity, and pay equity, offering transparency in labor practices and supporting better governance.
- The system also tracks performance against sustainability initiatives, such as decreasing annual energy consumption by 10% and reaching a recycling target of at least 95%.



RESEARCH FINDINGS: AI IN ESG REPORTING

- AI-driven ESG reporting tools automate data collection, analysis, and reporting, reducing human error.
- Real-time data integration enhances ESG reporting accuracy, enabling businesses to monitor performance consistently.
- AI-powered insights provide transparency in ESG performance, guiding organizations to make informed decisions.
- The implementation of AI improves compliance with ESG regulations and enhances organizational sustainability efforts.
- AI-based reporting tools provide businesses with the ability to meet complex regulatory standards efficiently.

KEY FINDINGS AND DISCUSSION

- AI-based tools have enhanced ESG transparency, compliance, and performance by automating data collection and report generation.
- Circular economy practices and waste management efficiency have improved with AI monitoring and reporting.
- AI tools monitor workforce diversity and gender balance, ensuring fair practices and equitable policies.
- Challenges like gender pay gaps and turnover rates highlight areas for further improvement and strategic action.
- AI transforms ESG reporting into a strategic tool for sustainable decision-making, rather than just a compliance task.

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Thank
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